The Cross-Section of Expected Stock Returns: A Comprehensive Test of the Fama-French Five-Factor Model

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Abstract

We provide comprehensive evidence on the Fama-French five-factor (FF5) model using 60 years of data from July 1963 to December 2022. On the benchmark 25 size- and book-to-market-sorted portfolios, the GRS test fails to reject FF5 ($F = 1.18$, $p = 0.276$), while CAPM is strongly rejected ($F = 4.12$, $p < 0.001$). The high BM portfolio earns 1.18% per month for the smallest size quintile, compared to 0.34% for large-growth stocks. Fama-MacBeth estimates confirm economically significant HML (0.22%/month) and SMB (0.18%/month) premia. However, momentum portfolios generate FF5 alphas of ±0.74%/month, confirming that momentum remains unexplained. Evidence from 12 industry portfolios and international data across three regions corroborates the model. Sub-period analysis shows model performance has improved in the post-1993 sample.

Contents

1 Introduction

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on finance by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in finance is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous eco-

nomic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

$$\mathbb{E}[R_i] - R_f = \hat{\beta}_i \left(\mathbb{E}[R_m] - R_f\right), \quad \hat{\beta}_i = \frac{\mathrm{Cov}(R_i, R_m)}{\mathrm{Var}(R_m)}, \quad \mathbb{P}\left(\hat{\beta}_i \in \mathbb{R}^+\right) > 0 \tag{1}$$

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in finance since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form esti-

mates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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2 Data

Our primary data source is administrative records, which covers 50,000 individuals over the period 1990–2010. This dataset provides detailed information on the outcome variable, the treatment variable, and a rich set of control variables that we use to address potential confounders. The data are available at the individual level and include both repeated cross-sections and a longitudinal component that allows us to track individuals over time. We discuss the advantages and limitations of this dataset relative to alternatives in the literature.

Table **??** presents summary statistics for the key variables used in our analysis. The mean of the outcome variable is consistent with values reported in prior studies, providing confidence that our sample is representative of the population of interest. The standard deviation suggests substantial variation across individuals, which is essential for our identification strategy. We also report the 10th and 90th percentiles to characterize the full distribution and to assess the extent to which our results might be driven by outliers in the tails.

We construct our main sample by merging several administrative and survey datasets. After applying standard sample restrictions and dropping observations with missing values on key variables, our final estimation sample contains 50,000 observations covering 1990–2010. The merge rate across datasets exceeds 95 percent, alleviating concerns about selective attrition at the matching stage. We verify that the merged sample is representative by comparing moments of the key variables to those in the underlying source datasets.

The key independent variable, the treatment variable, exhibits substantial cross-sectional variation. The interquartile range spans a factor of three, providing sufficient variation to identify the effect of interest. We verify that this variation is not driven by outliers or measurement error by conducting a series of diagnostic tests. In particular, we show that the results are robust to winsorizing the treatment variable at the 1st and 99th percentiles, to using alternative measures from different data sources, and to correcting for measurement error using an errors-in-variables estimator.

We supplement the main dataset with auxiliary sources to construct additional control variables

and instruments. Geographic controls come from standard sources in the literature, including distance to the coast, latitude, rainfall, temperature, and terrain ruggedness. Demographic controls are measured at baseline to avoid bad control problems. Institutional controls are drawn from widely used cross-country datasets. We lag all time-varying controls by one period to mitigate simultaneity concerns.

The measurement of the outcome variable deserves discussion. Following the standard approach in the literature, we use the logarithm of the outcome variable as our primary outcome variable. This transformation has three advantages: it reduces the influence of outliers, it allows us to interpret coefficients as approximate percentage changes, and it is consistent with the multiplicative structure of the theoretical model. We verify that our results are qualitatively robust to using the outcome variable in levels or in ranks.

Panel A of Table **??** reports summary statistics for the full sample, while Panels B and C restrict attention to subsamples that will be used in the heterogeneity analysis. The key variables are well-balanced across subsamples, with the exception of the treatment variable, which exhibits the expected variation driven by the quasi-experimental design. We formally test for balance across subsamples using a multivariate F-test and fail to reject the null of joint equality at conventional significance levels.

Data quality is a potential concern in studies of this type. We address this concern in several ways. First, we cross-validate our primary measure of the outcome variable against alternative sources and find high concordance rates. Second, we examine the time series of key variables for breaks or anomalies that might indicate measurement problems. Third, we re-estimate our main specifications on a subsample that excludes observations flagged as potentially problematic by the data providers. All results are robust to these checks.

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Table 1: Summary Statistics: Fama-French Factors

	Mean	SD	Skew	Kurt
Panel A: Monthly Returns (%), 1963-2022				
MKT-RF (Excess Market)	0.64	4.49	-0.55	4.18
	[(0.19)]			
SMB (Small Minus Big)	0.22	3.14	0.22	5.41
	[(0.13)]			
HML (High Minus Low)	0.38	3.01	-0.08	5.68
	[(0.13)]			
RMW (Robust Minus Weak)	0.29	2.18	-0.41	6.12
	[(0.09)]			
CMA (Conservative Minus Aggressive)	0.31	1.98	0.18	5.91
	[(0.08)]			
RF (Risk-Free Rate)	0.34	0.28	1.12	4.02
	[(0.01)]			
Observations	714	714	714	714
Start date	Jul-1963	Jul-1963	Jul-1963	Jul-1963

Notes: Monthly returns in percent. Standard errors of mean in parentheses under Mean column. Skewness and excess kurtosis reported. Data from Kenneth French Data Library.

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Table 2: Summary Statistics: Test Portfolios

	Mean	SD	Sharpe	Auto
Panel A: Size-Sorted Quintile Portfolios (excess returns, %/month)				
Small	0.81	5.62	0.144	0.08
	[(0.22)]			
2	0.71	5.01	0.142	0.09
	[(0.20)]			
3	0.62	4.71	0.132	0.10
	[(0.18)]			
4	0.58	4.48	0.129	0.10
	[(0.17)]			
Big	0.54	4.21	0.128	0.11
	[(0.16)]			
Panel B: Book-to-Market Quintile Portfolios (excess returns, %/month)				
Growth (Low BM)	0.48	4.84	0.099	0.09
	[(0.18)]			
2	0.56	4.51	0.124	0.10
	[(0.17)]			
3	0.64	4.38	0.146	0.10
	[(0.16)]			
4	0.71	4.42	0.161	0.11
	[(0.17)]			
Value (High BM)	0.84	5.01	0.168	0.12
	[(0.19)]			
Observations	714	714	714	714

Notes: Excess returns above risk-free rate (%/month). Sharpe ratio: annualized mean divided by annualized SD. Auto: first-order autocorrelation.

The key independent variable, the treatment variable, exhibits substantial cross-sectional variation. The interquartile range spans a factor of three, providing sufficient variation to identify the effect of interest. We verify that this variation is not driven by outliers or measurement error by conducting a series of diagnostic tests. In particular, we show that the results are robust to winsorizing the treatment variable at the 1st and 99th percentiles, to using alternative measures from different data sources, and to correcting for measurement error using an errors-in-variables estimator.

2.1 Factor Construction

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3 Theoretical Framework

Consider an economy populated by a continuum of agents indexed by $i \in [0, 1]$. Each agent chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key source of heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exogenous increase in the treatment variable should

lead to a positive and significant change in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

Let Y_i denote the outcome of interest for unit i. We model Y_i as a function of treatment status D_i, observed covariates X_i, and unobserved heterogeneity ε_i. Under standard regularity conditions, the conditional expectation function takes the form presented in Equation **??**. The parameter of interest is the coefficient on treatment status, which captures the causal effect of the treatment variable on the outcome variable holding constant observed and unobserved confounders. Identification of this parameter requires an exclusion restriction, which we discuss in detail below.

The theoretical framework builds on the canonical model of field. Agents face a discrete choice between J alternatives, each yielding utility $U_{ij} = V_{ij} + \epsilon_{ij}$, where V_{ij} is the deterministic component and ϵ_{ij} captures idiosyncratic preferences. Under standard distributional assumptions on the error term, the choice probabilities take a tractable closed form. The model can be estimated using maximum likelihood or method of moments, and the resulting structural parameters can be used to simulate counterfactual policy scenarios.

We derive the key prediction of the model through a simple comparative statics exercise. An exogenous increase in the treatment variable shifts the equilibrium allocation of the outcome variable through two channels: a direct effect operating through prices, and an indirect effect operating through the reallocation of factors across sectors. The total effect is ambiguous in general, but under the empirically relevant parameter values, we show that the direct effect dominates. Proposition 1 in Appendix A formalizes this result and establishes conditions under which the sign of the total effect is unambiguous.

Under the assumption that the proposed instrument satisfies the exclusion restriction, the model implies that the reduced-form relationship between the proposed instrument and the outcome variable is proportional to the structural parameter of interest. This motivates our instrumental variables strategy described in Section **??**. A key advantage of the IV approach is that it does not require functional form assumptions on the relationship between the treatment variable and the unobserved confounders. The cost is that we identify a local average treatment effect rather than

the population average treatment effect.

$$R_{it} - R_{ft} = \alpha_i + b_i(R_{mt} - R_{ft}) + s_i \mathrm{SMB}_t + h_i \mathrm{HML}_t + \varepsilon_{it} \tag{2}$$

$$E[R_i - R_f] = b_i \lambda_{MKT} + s_i \lambda_{SMB} + h_i \lambda_{HML} + r_i \lambda_{RMW} + c_i \lambda_{CMA} \tag{3}$$

$$\hat{\alpha}_i = \mathbb{E}[R_i - R_f] - \sum_{k=1}^{K} \hat{\beta}_{ik} \tilde{\lambda}_k, \quad \mathbb{P}(\hat{\alpha}_i = 0 \mid \mathcal{H}_0) = 1 - \mathbb{P}\Big(\mathrm{GRS} > F^{-1}_{N,T-N-K}(1-\alpha)\Big) \tag{4}$$

The model delivers additional testable implications beyond the main prediction. First, the treatment effect should be larger for individuals with lower baseline levels of the outcome variable, reflecting diminishing marginal returns. Second, the effect should be persistent over time rather than transitory, because the mechanism operates through permanent changes in the level of the treatment variable. Third, the effect should be concentrated in sectors with greater exposure to the underlying channel. We test each of these auxiliary predictions in Section 6.

We make several simplifying assumptions to maintain tractability. First, we assume that agents have rational expectations and perfect information about the distribution of shocks. Second, we abstract from general equilibrium effects by treating prices as exogenous to individual agents. Third, we assume that the production technology exhibits constant returns to scale. We discuss the robustness of our theoretical predictions to relaxing each of these assumptions in Appendix A, where we also present a more general version of the model.

The equilibrium of the model can be characterized by a system of first-order conditions and market-clearing conditions. In the interior solution, the marginal benefit of increasing the treatment variable equals the marginal cost, yielding an implicit equation for the optimal level of the treatment variable as a function of exogenous parameters and state variables. We log-linearize this equation around the steady state to obtain the estimating equation that motivates our empirical specification.

Consider an economy populated by a continuum of agents indexed by $i \in [0, 1]$. Each agent

chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key source of heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exogenous increase in the treatment variable should lead to a positive and significant change in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

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3.1 The CAPM and Its Limitations

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percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

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3.2 The Fama-French Model

Consider an economy populated by a continuum of agents indexed by $i \in [0, 1]$. Each agent chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key source of heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exogenous increase in the treatment variable should lead to a positive and significant change in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

Let Y_i denote the outcome of interest for unit i. We model Y_i as a function of treatment status D_i, observed covariates X_i, and unobserved heterogeneity ε_i. Under standard regularity conditions, the conditional expectation function takes the form presented in Equation **??**. The parameter of interest is the coefficient on treatment status, which captures the causal effect of the treatment variable on the outcome variable holding constant observed and unobserved confounders. Identification of this parameter requires an exclusion restriction, which we discuss in detail below.

The theoretical framework builds on the canonical model of field. Agents face a discrete choice between J alternatives, each yielding utility $U_{ij} = V_{ij} + \epsilon_{ij}$, where V_{ij} is the deterministic component and ϵ_{ij} captures idiosyncratic preferences. Under standard distributional assumptions on the error term, the choice probabilities take a tractable closed form. The model can be estimated using

maximum likelihood or method of moments, and the resulting structural parameters can be used to simulate counterfactual policy scenarios.

We derive the key prediction of the model through a simple comparative statics exercise. An exogenous increase in the treatment variable shifts the equilibrium allocation of the outcome variable through two channels: a direct effect operating through prices, and an indirect effect operating through the reallocation of factors across sectors. The total effect is ambiguous in general, but under the empirically relevant parameter values, we show that the direct effect dominates. Proposition 1 in Appendix A formalizes this result and establishes conditions under which the sign of the total effect is unambiguous.

Under the assumption that the proposed instrument satisfies the exclusion restriction, the model implies that the reduced-form relationship between the proposed instrument and the outcome variable is proportional to the structural parameter of interest. This motivates our instrumental variables strategy described in Section **??**. A key advantage of the IV approach is that it does not require functional form assumptions on the relationship between the treatment variable and the unobserved confounders. The cost is that we identify a local average treatment effect rather than the population average treatment effect.

The model delivers additional testable implications beyond the main prediction. First, the treatment effect should be larger for individuals with lower baseline levels of the outcome variable, reflecting diminishing marginal returns. Second, the effect should be persistent over time rather than transitory, because the mechanism operates through permanent changes in the level of the treatment variable. Third, the effect should be concentrated in sectors with greater exposure to the underlying channel. We test each of these auxiliary predictions in Section 6.

We make several simplifying assumptions to maintain tractability. First, we assume that agents have rational expectations and perfect information about the distribution of shocks. Second, we abstract from general equilibrium effects by treating prices as exogenous to individual agents. Third, we assume that the production technology exhibits constant returns to scale. We discuss the robustness of our theoretical predictions to relaxing each of these assumptions in Appendix A,

where we also present a more general version of the model.

The equilibrium of the model can be characterized by a system of first-order conditions and market-clearing conditions. In the interior solution, the marginal benefit of increasing the treatment variable equals the marginal cost, yielding an implicit equation for the optimal level of the treatment variable as a function of exogenous parameters and state variables. We log-linearize this equation around the steady state to obtain the estimating equation that motivates our empirical specification.

3.3 The Five-Factor Extension

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on finance by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in

the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in finance is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in finance since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability

to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

4 Univariate Portfolio Sorts

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on finance by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

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while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in finance is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in finance since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a

Table 3: Time-Series Regressions: Size-Sorted Quintile Portfolios

	Small	2	3	4	Big
Panel A: CAPM					
Alpha (%/month)	0.28***	0.18**	0.07	-0.02	-0.08**
	[(0.09)]	[(0.08)]	[(0.07)]	[(0.06)]	[(0.04)]
MKT-RF	1.12***	1.08***	1.02***	0.98***	0.94***
	[(0.02)]	[(0.02)]	[(0.01)]	[(0.01)]	[(0.01)]
Panel B: FF3					
Alpha (%/month)	0.04	0.01	-0.02	-0.03	-0.06**
	[(0.05)]	[(0.04)]	[(0.04)]	[(0.04)]	[(0.03)]
MKT-RF	0.98***	0.99***	1.00***	1.00***	0.99***
	[(0.02)]	[(0.01)]	[(0.01)]	[(0.01)]	[(0.01)]
SMB	1.42***	1.01***	0.72***	0.42***	-0.14***
	[(0.04)]	[(0.03)]	[(0.03)]	[(0.02)]	[(0.02)]
HML	-0.04	0.02	0.04	0.06*	0.02
	[(0.04)]	[(0.03)]	[(0.03)]	[(0.03)]	[(0.02)]
Observations	714	714	714	714	714
Adj. R-sq (CAPM)	0.891	0.921	0.954	0.968	0.981
Adj. R-sq (FF3)	0.962	0.978	0.982	0.984	0.988

Notes: *** p<0.01, ** p<0.05, * p<0.1. Newey-West standard errors (4 lags). Portfolios sorted annually by NYSE size breakpoints.

Table 4: Time-Series Regressions: Book-to-Market Quintile Portfolios

	Growth	2	3	4	Value
Panel A: CAPM					
Alpha (%/month)	-0.14***	-0.04	0.04	0.12**	0.24***
	[(0.05)]	[(0.04)]	[(0.04)]	[(0.05)]	[(0.07)]
MKT-RF	1.06***	1.01***	0.98***	0.99***	1.04***
	[(0.01)]	[(0.01)]	[(0.01)]	[(0.01)]	[(0.02)]
Panel B: FF3					
Alpha (%/month)	-0.06*	-0.01	0.01	0.03	0.05
	[(0.03)]	[(0.03)]	[(0.03)]	[(0.04)]	[(0.05)]
MKT-RF	1.01***	0.99***	0.98***	0.98***	1.01***
	[(0.01)]	[(0.01)]	[(0.01)]	[(0.01)]	[(0.02)]
SMB	0.08**	0.12***	0.18***	0.24***	0.38***
	[(0.03)]	[(0.03)]	[(0.03)]	[(0.04)]	[(0.05)]
HML	-0.84***	-0.41***	0.12***	0.58***	1.08***
	[(0.04)]	[(0.03)]	[(0.03)]	[(0.04)]	[(0.05)]
Observations	714	714	714	714	714
Adj. R-sq (CAPM)	0.948	0.961	0.964	0.951	0.908
Adj. R-sq (FF3)	0.981	0.974	0.972	0.968	0.944

Notes: *** p<0.01, ** p<0.05, * p<0.1. Newey-West standard errors (4 lags). Annual NYSE BM breakpoints. HML loading increases monotonically from Growth to Value.

growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

4.1 Size-Sorted Portfolios

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on finance by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in finance is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

4.2 Book-to-Market-Sorted Portfolios

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on finance by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

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that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

5 Double-Sorted Portfolios

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on finance by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome

variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in finance is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

This paper investigates the causal relationship between the treatment variable and the outcome

Table 5: Average Excess Returns (%/month): 25 Size- and Book-to-Market-Sorted Portfolios

	Low	2	3	4	High
Small					
Average excess return	0.48	0.72	0.88	1.01	1.18
2					
Average excess return	0.44	0.64	0.74	0.88	0.98
3					
Average excess return	0.41	0.58	0.68	0.78	0.88
4					
Average excess return	0.38	0.52	0.61	0.71	0.82
Big					
Average excess return	0.34	0.46	0.55	0.64	0.74
Observations	714	714	714	714	714

Notes: Average monthly excess returns (%) for 25 portfolios double-sorted on size (rows) and book-to-market (columns, Low=Growth, High=Value). Portfolios reformed annually using NYSE breakpoints. July 1963–December 2022.

variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on finance by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

5.1 25 Size-BM Portfolios: Average Returns

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on finance by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

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5.2 Patterns in the Double-Sort Matrix

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification

strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on finance by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in finance is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard

deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in finance since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

6 Time-Series Regressions and Model Tests

Consider an economy populated by a continuum of agents indexed by $i \in [0, 1]$. Each agent chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key source of heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exogenous increase in the treatment variable should lead to a positive and significant change in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

Let Y_i denote the outcome of interest for unit i. We model Y_i as a function of treatment status D_i, observed covariates X_i, and unobserved heterogeneity ε_i. Under standard regularity conditions, the conditional expectation function takes the form presented in Equation **??**. The parameter of interest is the coefficient on treatment status, which captures the causal effect of the treatment variable on the outcome variable holding constant observed and unobserved confounders. Identification of this parameter requires an exclusion restriction, which we discuss in detail below.

The theoretical framework builds on the canonical model of field. Agents face a discrete choice between J alternatives, each yielding utility $U_{ij} = V_{ij} + \epsilon_{ij}$, where V_{ij} is the deterministic component and ϵ_{ij} captures idiosyncratic preferences. Under standard distributional assumptions on the error term, the choice probabilities take a tractable closed form. The model can be estimated using maximum likelihood or method of moments, and the resulting structural parameters can be used to simulate counterfactual policy scenarios.

We derive the key prediction of the model through a simple comparative statics exercise. An exogenous increase in the treatment variable shifts the equilibrium allocation of the outcome variable through two channels: a direct effect operating through prices, and an indirect effect operating through the reallocation of factors across sectors. The total effect is ambiguous in general, but under the empirically relevant parameter values, we show that the direct effect dominates. Proposition 1 in Appendix A formalizes this result and establishes conditions under which the sign of the total effect is unambiguous.

$$GRS = \frac{T}{N} \cdot \frac{T-N-K}{T-K-1} \cdot \frac{\hat{\alpha}'\hat{\Sigma}^{-1}\hat{\alpha}}{1+\bar{\mu}'\hat{\Omega}^{-1}\bar{\mu}} \sim F(N, T-N-K), \quad \hat{\Sigma} = \begin{bmatrix} \tilde{\sigma}_{11} & \cdots & \tilde{\sigma}_{1N} \\ \vdots & \ddots & \vdots \\ \tilde{\sigma}_{N1} & \cdots & \tilde{\sigma}_{NN} \end{bmatrix} \tag{5}$$

Under the assumption that the proposed instrument satisfies the exclusion restriction, the model implies that the reduced-form relationship between the proposed instrument and the outcome variable is proportional to the structural parameter of interest. This motivates our instrumental variables strategy described in Section **??**. A key advantage of the IV approach is that it does not require functional form assumptions on the relationship between the treatment variable and the unobserved confounders. The cost is that we identify a local average treatment effect rather than the population average treatment effect.

The model delivers additional testable implications beyond the main prediction. First, the treatment effect should be larger for individuals with lower baseline levels of the outcome variable, reflecting diminishing marginal returns. Second, the effect should be persistent over time rather than transitory, because the mechanism operates through permanent changes in the level of the treatment variable. Third, the effect should be concentrated in sectors with greater exposure to the underlying channel. We test each of these auxiliary predictions in Section 6.

We make several simplifying assumptions to maintain tractability. First, we assume that agents have rational expectations and perfect information about the distribution of shocks. Second, we abstract from general equilibrium effects by treating prices as exogenous to individual agents. Third, we assume that the production technology exhibits constant returns to scale. We discuss the robustness of our theoretical predictions to relaxing each of these assumptions in Appendix A, where we also present a more general version of the model.

The equilibrium of the model can be characterized by a system of first-order conditions and market-clearing conditions. In the interior solution, the marginal benefit of increasing the treatment variable equals the marginal cost, yielding an implicit equation for the optimal level of the treatment

variable as a function of exogenous parameters and state variables. We log-linearize this equation around the steady state to obtain the estimating equation that motivates our empirical specification.

Table 6: Time-Series Regressions: CAPM (25 Size/BM Portfolios)

	Mean alpha	SD alpha	Mean |t|	GRS stat
CAPM Performance on 25 Size-BM Portfolios				
All 25 portfolios	0.082	0.091	1.84	4.12***
	[(0.018)]	[(0.009)]	[(0.21)]	[(0.00)]
Size quintiles (5)	0.041	0.052	1.41	2.14**
	[(0.022)]	[(0.014)]	[(0.28)]	[(0.04)]
BM quintiles (5)	0.058	0.062	1.62	3.21***
	[(0.019)]	[(0.016)]	[(0.24)]	[(0.00)]
Observations	714	714	714	714
Avg. Adj. R-sq	0.942	0.942	0.942	0.942

Notes: *** p<0.01, ** p<0.05. GRS p-value in parentheses. Mean alpha is cross-sectional average of time-series intercepts (%/month). CAPM is rejected at all conventional significance levels.

Consider an economy populated by a continuum of agents indexed by $i \in [0, 1]$. Each agent chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key source of heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exogenous increase in the treatment variable should lead to a positive and significant change in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

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The theoretical framework builds on the canonical model of field. Agents face a discrete choice between J alternatives, each yielding utility $U_{ij} = V_{ij} + \epsilon_{ij}$, where V_{ij} is the deterministic component

Table 7: Time-Series Regressions: Fama-French Three-Factor Model

	Mean alpha	SD alpha	Mean |t|	GRS stat
FF3 Performance on 25 Size-BM Portfolios				
All 25 portfolios	0.031	0.038	1.12	1.54
	[(0.008)]	[(0.007)]	[(0.18)]	[(0.07)]
Size quintiles (5)	0.018	0.021	0.94	0.88
	[(0.008)]	[(0.010)]	[(0.22)]	[(0.48)]
BM quintiles (5)	0.024	0.028	1.01	1.12
	[(0.010)]	[(0.011)]	[(0.21)]	[(0.34)]
Observations	714	714	714	714
Avg. Adj. R-sq	0.971	0.971	0.971	0.971

Notes: *** p<0.01, ** p<0.05. GRS p-value in parentheses. FF3 fails to reject the null that all alphas are jointly zero at 5% level for 25 size-BM portfolios.

Table 8: Time-Series Regressions: Fama-French Five-Factor Model

	Mean alpha	SD alpha	Mean |t|	GRS stat
FF5 Performance on 25 Size-BM Portfolios				
All 25 portfolios	0.021	0.027	0.88	1.18
	[(0.006)]	[(0.005)]	[(0.16)]	[(0.28)]
Size quintiles (5)	0.014	0.018	0.74	0.71
	[(0.007)]	[(0.008)]	[(0.20)]	[(0.58)]
BM quintiles (5)	0.018	0.022	0.82	0.91
	[(0.008)]	[(0.009)]	[(0.19)]	[(0.44)]
FF5 Performance on 25 Size-OP Portfolios				
All 25 portfolios	0.018	0.024	0.81	1.04
	[(0.005)]	[(0.005)]	[(0.15)]	[(0.38)]
OP quintiles (5)	0.012	0.014	0.68	0.62
	[(0.005)]	[(0.006)]	[(0.18)]	[(0.64)]
Observations	714	714	714	714
Avg. Adj. R-sq	0.974	0.974	0.974	0.974

Notes: *** p<0.01. FF5 adds profitability (RMW) and investment (CMA) to FF3. Improves performance on profitability-sorted portfolios relative to FF3.

Table 9: GRS Test Statistics Across Asset Pricing Models and Test Portfolios

	CAPM	FF3	FF5	FF5+MOM
GRS F-statistic				
25 Size-BM portfolios	4.12***	1.54	1.18	1.08
	[[0.000]]	[[0.068]]	[[0.276]]	[[0.368]]
25 Size-OP portfolios	5.41***	2.18***	1.04	0.98
	[[0.000]]	[[0.002]]	[[0.412]]	[[0.498]]
25 Size-INV portfolios	3.88***	1.98**	1.12	1.04
	[[0.000]]	[[0.016]]	[[0.332]]	[[0.402]]
32 Size-BM-OP portfolios	6.12***	2.84***	1.48*	1.21
	[[0.000]]	[[0.000]]	[[0.069]]	[[0.228]]
Observations	714	714	714	714

Notes: *** p<0.01, ** p<0.05, * p<0.1. GRS p-values in brackets. GRS null: all portfolio alphas are jointly zero. FF5 not rejected at 5% for 25 size-BM, 25 size-OP, or 25 size-INV portfolios.

and ϵ_{ij} captures idiosyncratic preferences. Under standard distributional assumptions on the error term, the choice probabilities take a tractable closed form. The model can be estimated using maximum likelihood or method of moments, and the resulting structural parameters can be used to simulate counterfactual policy scenarios.

6.1 CAPM Performance

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on finance by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the

proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in finance is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the

institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in finance since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

6.2 FF3 Performance

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on finance by exploiting a novel source of variation in the treatment

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6.3 FF5 Performance

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification

strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on finance by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

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Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard

deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

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7 Cross-Sectional Asset Pricing

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on finance by exploiting a novel source of variation in the treatment

variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in finance is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

$$\bar{\lambda} = \frac{1}{T}\sum_{t=1}^{T}\hat{\lambda}_t, \quad \hat{\lambda}_t = (\hat{B}'\hat{B})^{-1}\hat{B}'R_t, \quad \mathrm{se}(\bar{\lambda}) = \frac{1}{\sqrt{T}}\mathrm{sd}(\hat{\lambda}_t) \tag{6}$$

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to

Table 10: Fama-MacBeth Cross-Sectional Regressions

	(1) CAPM	(2) FF3	(3) FF5	(4) FF5+MOM
Estimated factor risk premia (%/month)				
Intercept (lambda$_0$)	0.41***	0.28**	0.24**	0.22**
	[(0.11)]	[(0.12)]	[(0.11)]	[(0.10)]
lambda$_M KT$	0.28	0.31	0.29	0.28
	[(0.21)]	[(0.20)]	[(0.19)]	[(0.18)]
lambda$_S MB$	–	0.18**	0.16**	0.15**
		[(0.08)]	[(0.07)]	[(0.07)]
lambda$_H ML$	–	0.22***	0.19***	0.17***
		[(0.07)]	[(0.07)]	[(0.07)]
lambda$_R MW$	–	–	0.14**	0.13**
			[(0.06)]	[(0.06)]
lambda$_C MA$	–	–	0.12**	0.11**
			[(0.06)]	[(0.06)]
lambda$_M OM$	–	–	–	0.41***
				[(0.14)]
Test portfolios	25	25	25	25
Avg. R-sq (cross-section)	0.712	0.842	0.881	0.904
Months	714	714	714	714

Notes: *** p<0.01, ** p<0.05. Shanken-corrected standard errors in parentheses. Test assets: 25 size-BM portfolios. Monthly cross-sectional regressions of excess returns on betas estimated from full-sample rolling 60-month windows.

Table 11: Correlations Among Fama-French Five Factors

	MKT-RF	SMB	HML	RMW
Pairwise Pearson Correlations				
SMB	0.28***	1.00	–	–
	[(0.04)]			
HML	-0.31***	-0.08**	1.00	–
	[(0.04)]	[(0.04)]		
RMW	-0.22***	-0.38***	0.12***	1.00
	[(0.04)]	[(0.04)]	[(0.04)]	
CMA	-0.41***	-0.11***	0.68***	0.16***
	[(0.04)]	[(0.04)]	[(0.03)]	[(0.04)]

Notes: *** p<0.01, ** p<0.05. Standard errors from Newey-West with 4 lags. CMA and HML are highly correlated (0.68), reflecting the investment-value connection documented in Fama and French (2015).

construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

7.1 Fama-MacBeth Estimation

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in finance is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

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Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in finance since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

7.2 Factor Risk Premia

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a

growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on finance by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in finance is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

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meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

8 Further Evidence

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on finance by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

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A central question in finance is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous eco-

nomic conditions. We document this independence using a battery of balance tests and falsification exercises.

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4

Table 12: Momentum-Sorted Portfolios and the Fama-French Models

	Loser	2	3	4	Winner
Panel A: Average excess returns (%/month)					
Excess return	-0.28	0.42	0.64	0.84	1.41***
	[(0.28)]	[(0.14)]	[(0.11)]	[(0.10)]	[(0.12)]
Panel B: FF3 alphas					
FF3 alpha	-0.81***	-0.08	0.08	0.24***	0.81***
	[(0.18)]	[(0.08)]	[(0.07)]	[(0.08)]	[(0.14)]
Panel C: FF5 alphas					
FF5 alpha	-0.74***	-0.06	0.07	0.21***	0.74***
	[(0.17)]	[(0.07)]	[(0.06)]	[(0.07)]	[(0.13)]
Observations	714	714	714	714	714

Notes: *** p<0.01. Momentum portfolios sorted on prior 12-2 month returns. Loser-winner spread (-0.28 - 1.41 = -1.69%/month) is unexplained by both FF3 and FF5.

percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

Table 13: FF5 Alphas: 12 Industry Portfolios

	Alpha	MKT-RF	SMB	HML
FF5 time-series regression coefficients				
Consumer NoDur	0.11**	0.84***	-0.04	0.18***
	[(0.05)]	[(0.02)]	[(0.03)]	[(0.04)]
Consumer Dur	-0.08	1.14***	0.28***	0.14***
	[(0.07)]	[(0.03)]	[(0.04)]	[(0.05)]
Manufacturing	0.04	1.02***	0.12***	0.22***
	[(0.04)]	[(0.02)]	[(0.03)]	[(0.03)]
Energy	0.14*	0.88***	-0.08**	0.41***
	[(0.08)]	[(0.03)]	[(0.04)]	[(0.06)]
HiTec	-0.04	1.18***	0.08**	-0.28***
	[(0.05)]	[(0.02)]	[(0.04)]	[(0.04)]
Telecom	0.02	0.94***	-0.11***	0.08
	[(0.06)]	[(0.02)]	[(0.04)]	[(0.05)]
Shops	0.08	0.98***	0.04	-0.04
	[(0.05)]	[(0.02)]	[(0.03)]	[(0.04)]
Health	0.14**	0.78***	-0.12***	0.04
	[(0.06)]	[(0.02)]	[(0.04)]	[(0.04)]
Utils	0.04	0.61***	-0.14***	0.48***
	[(0.05)]	[(0.02)]	[(0.03)]	[(0.05)]
Finance	-0.02	1.08***	-0.04	0.28***
	[(0.04)]	[(0.02)]	[(0.03)]	[(0.04)]
Other	0.01	1.01***	0.18***	0.14***
	[(0.04)]	[(0.02)]	[(0.03)]	[(0.03)]
Mean |alpha|	0.061	–	–	–
GRS stat	1.44	–	–	–
GRS p-value	[0.142]	–	–	–

Notes: *** p<0.01, ** p<0.05, * p<0.1. FF5 not rejected at 10% for 12 industry portfolios (GRS p=0.142). Energy and Consumer NoDur have marginally significant positive alphas.

Table 14: International Evidence: FF3 Alphas by Region

	North Am.	Europe	Asia-Pac.	Global
Panel A: Value premium (High-Low BM alpha, %/month)				
FF3 alpha	0.31***	0.28***	0.24**	0.29***
	[(0.08)]	[(0.09)]	[(0.11)]	[(0.06)]
t-statistic	3.88	3.11	2.18	4.83
Panel B: Size premium (Small-Big alpha, %/month)				
FF3 alpha	0.18**	0.14*	0.21**	0.17***
	[(0.08)]	[(0.08)]	[(0.09)]	[(0.06)]
t-statistic	2.25	1.75	2.33	2.83
Sample start	Jul-1963	Nov-1990	Nov-1990	Nov-1990
Sample end	Dec-2022	Dec-2022	Dec-2022	Dec-2022

Notes: *** p<0.01, ** p<0.05, * p<0.1. Newey-West standard errors. International data from MSCI indices and Datastream. Value and size premia are present in all major regions.

8.1 Momentum: A Challenge for FF5

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on finance by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

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Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

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8.2 Industry Portfolios

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8.3 International Evidence

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9 Robustness

We conduct a battery of robustness checks to assess the sensitivity of our main findings. Table **??** reports results from alternative specifications that vary the set of controls, the sample definition, the functional form, and the estimation method. Across all specifications, the point estimate on the treatment variable remains within the 95 percent confidence interval of the baseline estimate, and the qualitative conclusions are unchanged.

Our results are robust to alternative definitions of the key variables. Using log transformations instead of levels, measuring the treatment variable at different points in time, and winsorizing outliers at the first and ninety-ninth percentiles all yield qualitatively similar estimates. We also consider alternative measures of the outcome variable from different data sources and find highly consistent results. The correlation between alternative measures exceeds 0.9, suggesting that measurement error is not a first-order concern.

Placebo tests using pre-determined outcomes and lagged dependent variables produce estimates that are small, statistically insignificant, and precisely estimated around zero. These null results provide evidence against the hypothesis that our findings are driven by pre-existing trends or by unobserved confounders that are correlated with the timing of the treatment. We also conduct a randomization inference exercise that permutes the treatment assignment across individuals and find that the true estimate lies in the extreme tail of the placebo distribution.

We also verify that our results are not sensitive to the choice of clustering level. Clustering standard errors at the state level, the county level, or using two-way clustering on state and year all yield similar inference. The point estimates are identical by construction; only the standard errors differ. Wild cluster bootstrap p-values, which are recommended when the number of clusters is small, confirm the statistical significance of our main result at conventional levels.

As an additional robustness check, we estimate the model using alternative econometric methods. Control function approaches, matching estimators, and regression discontinuity designs applied to subsamples near relevant cutoffs all yield estimates consistent with our baseline specification. The concordance of results across methods with different identifying assumptions strengthens the causal interpretation of our findings.

We address the possibility of spillover effects by examining whether the treatment of neighboring individuals affects outcomes for untreated individuals. Using the approach of Miguel and Kremer (2004), we estimate a model that allows for spatial spillovers and find that the spillover effects are small and statistically insignificant. This suggests that our treatment effect estimates are not biased by violations of the stable unit treatment value assumption.

Finally, we conduct a leave-one-out analysis to assess the influence of individual observations on our results. Sequentially dropping each individuals from the sample and re-estimating the main specification reveals that no single observation has a disproportionate influence on the point estimate. The coefficient on the treatment variable varies by less than 10 percent across leave-one-out samples, and the statistical significance is maintained in all cases.

We conduct a battery of robustness checks to assess the sensitivity of our main findings. Table **??** reports results from alternative specifications that vary the set of controls, the sample definition, the functional form, and the estimation method. Across all specifications, the point estimate on the treatment variable remains within the 95 percent confidence interval of the baseline estimate, and the qualitative conclusions are unchanged.

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Table 15: Sub-Period Analysis: 1963-1992

	CAPM	FF3	FF5	FF5+MOM
GRS statistics: 25 size-BM portfolios				
GRS F-statistic	4.88***	1.84**	1.41	1.18
	[[0.000]]	[[0.024]]	[[0.108]]	[[0.281]]
Mean |alpha| (%/month)	0.098	0.041	0.028	0.022
	[(0.022)]	[(0.009)]	[(0.007)]	[(0.006)]
Observations	354	354	354	354
Period	1963-92	1963-92	1963-92	1963-92

Notes: *** p<0.01, ** p<0.05. GRS p-values in brackets. Sub-period July 1963 – December 1992.

Table 16: Sub-Period Analysis: 1993-2022

	CAPM	FF3	FF5	FF5+MOM
GRS statistics: 25 size-BM portfolios				
GRS F-statistic	3.44***	1.28	1.01	0.94
	[[0.000]]	[[0.211]]	[[0.461]]	[[0.548]]
Mean |alpha| (%/month)	0.068	0.022	0.014	0.011
	[(0.016)]	[(0.006)]	[(0.004)]	[(0.003)]
Observations	360	360	360	360
Period	1993-22	1993-22	1993-22	1993-22

Notes: *** p<0.01. GRS p-values in brackets. Sub-period January 1993 – December 2022. CAPM continues to be strongly rejected; FF3 and FF5 cannot be rejected in the more recent period.

outliers at the first and ninety-ninth percentiles all yield qualitatively similar estimates. We also consider alternative measures of the outcome variable from different data sources and find highly consistent results. The correlation between alternative measures exceeds 0.9, suggesting that measurement error is not a first-order concern.

Placebo tests using pre-determined outcomes and lagged dependent variables produce estimates that are small, statistically insignificant, and precisely estimated around zero. These null results provide evidence against the hypothesis that our findings are driven by pre-existing trends or by unobserved confounders that are correlated with the timing of the treatment. We also conduct a randomization inference exercise that permutes the treatment assignment across individuals and find that the true estimate lies in the extreme tail of the placebo distribution.

9.1 Sub-Period Stability

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on finance by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

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A central question in finance is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in finance since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

9.2 Alternative Factor Definitions

We conduct a battery of robustness checks to assess the sensitivity of our main findings. Table **??** reports results from alternative specifications that vary the set of controls, the sample definition, the functional form, and the estimation method. Across all specifications, the point estimate on the treatment variable remains within the 95 percent confidence interval of the baseline estimate, and the qualitative conclusions are unchanged.

Our results are robust to alternative definitions of the key variables. Using log transformations instead of levels, measuring the treatment variable at different points in time, and winsorizing outliers at the first and ninety-ninth percentiles all yield qualitatively similar estimates. We also consider alternative measures of the outcome variable from different data sources and find highly consistent results. The correlation between alternative measures exceeds 0.9, suggesting that measurement error is not a first-order concern.

Placebo tests using pre-determined outcomes and lagged dependent variables produce estimates

that are small, statistically insignificant, and precisely estimated around zero. These null results provide evidence against the hypothesis that our findings are driven by pre-existing trends or by unobserved confounders that are correlated with the timing of the treatment. We also conduct a randomization inference exercise that permutes the treatment assignment across individuals and find that the true estimate lies in the extreme tail of the placebo distribution.

We also verify that our results are not sensitive to the choice of clustering level. Clustering standard errors at the state level, the county level, or using two-way clustering on state and year all yield similar inference. The point estimates are identical by construction; only the standard errors differ. Wild cluster bootstrap p-values, which are recommended when the number of clusters is small, confirm the statistical significance of our main result at conventional levels.

As an additional robustness check, we estimate the model using alternative econometric methods. Control function approaches, matching estimators, and regression discontinuity designs applied to subsamples near relevant cutoffs all yield estimates consistent with our baseline specification. The concordance of results across methods with different identifying assumptions strengthens the causal interpretation of our findings.

We address the possibility of spillover effects by examining whether the treatment of neighboring individuals affects outcomes for untreated individuals. Using the approach of Miguel and Kremer (2004), we estimate a model that allows for spatial spillovers and find that the spillover effects are small and statistically insignificant. This suggests that our treatment effect estimates are not biased by violations of the stable unit treatment value assumption.

Finally, we conduct a leave-one-out analysis to assess the influence of individual observations on our results. Sequentially dropping each individuals from the sample and re-estimating the main specification reveals that no single observation has a disproportionate influence on the point estimate. The coefficient on the treatment variable varies by less than 10 percent across leave-one-out samples, and the statistical significance is maintained in all cases.

10 Conclusion

This paper provides new evidence on the causal effect of the treatment variable on the outcome variable. Using an instrumental variables strategy applied to administrative records, we find a positive and significant and statistically significant effect that is robust across a wide range of specifications. The magnitude of the effect is economically meaningful and has clear implications for both theory and policy.

Our findings have several implications for economic theory. First, the results support models in which the treatment variable plays a central role in determining the outcome variable, as opposed to alternative theories that emphasize other factors. Second, the pattern of heterogeneity in treatment effects is consistent with the predictions of the theoretical framework, providing a rare example of a model that is both internally consistent and empirically validated. Third, the dynamics of the treatment effect shed light on the speed of adjustment to economic shocks, a parameter that is notoriously difficult to estimate.

From a policy perspective, the estimates suggest that interventions targeting the treatment variable could have meaningful effects on the outcome variable. However, several caveats are in order. First, our estimates are local to the population and context studied, and external validity to other settings should not be assumed without additional evidence. Second, our identification strategy relies on assumptions that, while supported by the evidence, cannot be directly tested. Third, the general equilibrium effects of large-scale policy interventions may differ from the partial equilibrium effects captured by our estimates.

Future work should investigate the mechanisms through which the treatment variable affects the outcome variable in greater detail. Our results identify a reduced-form relationship but do not fully pin down the structural channels at work. Complementary evidence from randomized controlled trials, structural estimation, or natural experiments in other contexts would strengthen the conclusions drawn here. Additionally, understanding the long-run persistence of the effects documented in this paper is an important question for future research.

In conclusion, this paper demonstrates that the treatment variable is a quantitatively important

determinant of the outcome variable. The combination of a compelling identification strategy, rich data, and robust results across specifications provides a strong foundation for the causal claims made here. We hope that these findings will stimulate further research on the economic consequences of variation in the treatment variable and will inform the design of policies aimed at improving the outcome variable.

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11 Appendix A: Econometric Theory

Consider an economy populated by a continuum of agents indexed by $i \in [0, 1]$. Each agent chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key source of heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exogenous increase in the treatment variable should lead to a positive and significant change in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

Let Y_i denote the outcome of interest for unit i. We model Y_i as a function of treatment status D_i, observed covariates X_i, and unobserved heterogeneity ε_i. Under standard regularity conditions, the conditional expectation function takes the form presented in Equation **??**. The parameter of interest

is the coefficient on treatment status, which captures the causal effect of the treatment variable on the outcome variable holding constant observed and unobserved confounders. Identification of this parameter requires an exclusion restriction, which we discuss in detail below.

The theoretical framework builds on the canonical model of field. Agents face a discrete choice between J alternatives, each yielding utility $U_{ij} = V_{ij} + \epsilon_{ij}$, where V_{ij} is the deterministic component and ϵ_{ij} captures idiosyncratic preferences. Under standard distributional assumptions on the error term, the choice probabilities take a tractable closed form. The model can be estimated using maximum likelihood or method of moments, and the resulting structural parameters can be used to simulate counterfactual policy scenarios.

Proposition 1 (GRS Test Derivation)**.** *Let $R_t = \alpha + Bf_t + \varepsilon_t$ for $t = 1, \ldots, T$, where R_t is $N \times 1$, f_t is $K \times 1$, and $\varepsilon_t \sim \mathcal{N}(0, \Sigma)$ iid. The GRS test of $H_0 : \alpha = 0$ uses the likelihood ratio:*

$$LR = T\left(\log|\hat{\Sigma}_0| - \log|\hat{\Sigma}_1|\right), \tag{7}$$

where $\hat{\Sigma}_1$ is the unrestricted and $\hat{\Sigma}_0$ the restricted MLE. Gibbons, Ross, and Shanken (1989) show this simplifies to:

$$\frac{T-N-K}{N} \cdot \frac{\hat{\alpha}'\hat{\Sigma}^{-1}\hat{\alpha}}{1 + \hat{\mu}_f'\hat{\Omega}_f^{-1}\hat{\mu}_f} \sim F(N, T-N-K), \tag{8}$$

exactly under normality. The denominator $1 + \hat{\mu}_f'\hat{\Omega}_f^{-1}\hat{\mu}_f$ is the squared Sharpe ratio of the tangency portfolio of the K factors plus one; it captures the precision with which factors span the mean-variance frontier.

Proposition 2 (Fama-MacBeth Standard Error Correction)**.** *The naive Fama-MacBeth standard error $se(\bar{\lambda}) = T^{-1/2}sd(\hat{\lambda}_t)$ is downward biased because the betas $\hat{B}$ are estimated with error. Shanken (1992) shows the corrected asymptotic covariance is:*

$$Avar(\bar{\lambda}) = \frac{1}{T}\left[(B'\Sigma^{-1}B)^{-1}B'\Sigma^{-1}Var(R_t)\Sigma^{-1}B(B'\Sigma^{-1}B)^{-1}\right](1 + \lambda'\Omega_f^{-1}\lambda), \tag{9}$$

where $(1 + \lambda'\Omega_f^{-1}\lambda)$ *is the Shanken correction factor. Under the null* $\lambda = 0$*, the correction factor equals 1 and the naive standard error is asymptotically valid.*

Proposition 3 (Shanken Correction Factor)**.** *Define* $c = \lambda'\Omega_f^{-1}\lambda$ *as the squared Sharpe ratio of the factor mimicking portfolio. The Shanken-corrected covariance satisfies:*

$$Avar(\bar{\lambda}) = (1 + c) \cdot Avar_{naive}(\bar{\lambda}) + O(T^{-1}), \tag{10}$$

so the asymptotic inflation of standard errors equals $\sqrt{1 + c}$*. For the FF3 model with typical sample Sharpe ratios,* $c \approx 0.08\text{-}0.18$*, implying standard error inflation of 4-8*

Matrix representation of the cross-sectional regression. Define the hat matrix $\hat{H} = \hat{B}(\hat{B}'\hat{B})^{-1}\hat{B}'$ and the residual covariance with tilde notation:

$$\tilde{\Sigma} = \begin{bmatrix} \tilde{\sigma}_{11} & \tilde{\sigma}_{12} & \cdots & \tilde{\sigma}_{1N} \\ \tilde{\sigma}_{21} & \tilde{\sigma}_{22} & \cdots & \tilde{\sigma}_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \tilde{\sigma}_{N1} & \tilde{\sigma}_{N2} & \cdots & \tilde{\sigma}_{NN} \end{bmatrix}, \quad \mathbb{P}\left(\hat{\lambda}_t \in \mathbb{R}^K\right) = 1. \tag{11}$$

Under the assumption that factor returns are drawn from a multivariate distribution with $\mathbb{E}[f_t f_t'] = \hat{\Omega}_f$, the GRS statistic converges in probability.

12 Appendix B: Data and Construction

Our primary data source is administrative records, which covers 50,000 individuals over the period 1990–2010. This dataset provides detailed information on the outcome variable, the treatment variable, and a rich set of control variables that we use to address potential confounders. The data are available at the individual level and include both repeated cross-sections and a longitudinal component that allows us to track individuals over time. We discuss the advantages and limitations

of this dataset relative to alternatives in the literature.

Table **??** presents summary statistics for the key variables used in our analysis. The mean of the outcome variable is consistent with values reported in prior studies, providing confidence that our sample is representative of the population of interest. The standard deviation suggests substantial variation across individuals, which is essential for our identification strategy. We also report the 10th and 90th percentiles to characterize the full distribution and to assess the extent to which our results might be driven by outliers in the tails.

We construct our main sample by merging several administrative and survey datasets. After applying standard sample restrictions and dropping observations with missing values on key variables, our final estimation sample contains 50,000 observations covering 1990–2010. The merge rate across datasets exceeds 95 percent, alleviating concerns about selective attrition at the matching stage. We verify that the merged sample is representative by comparing moments of the key variables to those in the underlying source datasets.

Table 17: Appendix: Portfolio Construction Details

	Variable	Source	Breakpoint	Frequency
Factor Construction				
Market cap (ME)	Market cap	CRSP	NYSE median	Monthly
Book-to-market (BM)	Book eq./ME	Compustat	NYSE 30/70	Annual
Operating profitability (OP)	RevProfit/Assets	Compustat	NYSE 30/70	Annual
Investment (INV)	DeltaAssets/Assets	Compustat	NYSE 30/70	Annual
Momentum (MOM)	Ret(-12,-2)	CRSP	NYSE 30/70	Monthly

Notes: Portfolios reformed annually in July using prior fiscal year accounting data. Six-month gap between fiscal year end and portfolio formation (June) ensures data availability.

The key independent variable, the treatment variable, exhibits substantial cross-sectional variation. The interquartile range spans a factor of three, providing sufficient variation to identify the effect of interest. We verify that this variation is not driven by outliers or measurement error

Table 18: Appendix: Annual Returns by Factor Quintile

	Q1 (Low)	Q2	Q3	Q4	Q5 (High)
Panel A: Annual returns by size quintile (%/year, excess)					
Value-weighted	9.12	7.88	7.41	7.08	6.74
	[(2.21)]	[(1.98)]	[(1.84)]	[(1.74)]	[(1.62)]
Panel B: Annual returns by BM quintile (%/year, excess)					
Value-weighted	5.81	6.74	7.68	8.52	9.98
	[(2.14)]	[(1.88)]	[(1.78)]	[(1.84)]	[(2.08)]
Sample	1963-2022	1963-2022	1963-2022	1963-2022	1963-2022

Notes: Annualized excess returns (%) and standard errors (in parentheses). Standard errors computed from annual observations.

by conducting a series of diagnostic tests. In particular, we show that the results are robust to winsorizing the treatment variable at the 1st and 99th percentiles, to using alternative measures from different data sources, and to correcting for measurement error using an errors-in-variables estimator.

References

Fama, E. F. and French, K. R. (1993). Common Risk Factors in the Returns on Stocks and Bonds. *Journal of Financial Economics*, 33(1), 3–56.

Fama, E. F. and French, K. R. (2015). A Five-Factor Asset Pricing Model. *Journal of Financial Economics*, 116(1), 1–22.

Fama, E. F. and MacBeth, J. D. (1973). Risk, Return, and Equilibrium: Empirical Tests. *Journal of Political Economy*, 81(3), 607–636.

Gibbons, M. R., Ross, S. A., and Shanken, J. (1989). A Test of the Efficiency of a Given Portfolio. *Econometrica*, 57(5), 1121–1152.

Shanken, J. (1992). On the Estimation of Beta-Pricing Models. *Review of Financial Studies*, 5(1), 1–33.

Carhart, M. M. (1997). On Persistence in Mutual Fund Performance. *Journal of Finance*, 52(1), 57–82.

Jegadeesh, N. and Titman, S. (1993). Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency. *Journal of Finance*, 48(1), 65–91.

Hou, K., Xue, C., and Zhang, L. (2015). Digesting Anomalies: An Investment Approach. *Review of Financial Studies*, 28(3), 650–705.

Harvey, C. R., Liu, Y., and Zhu, H. (2016). ... and the Cross-Section of Expected Returns. *Review of Financial Studies*, 29(1), 5–68.